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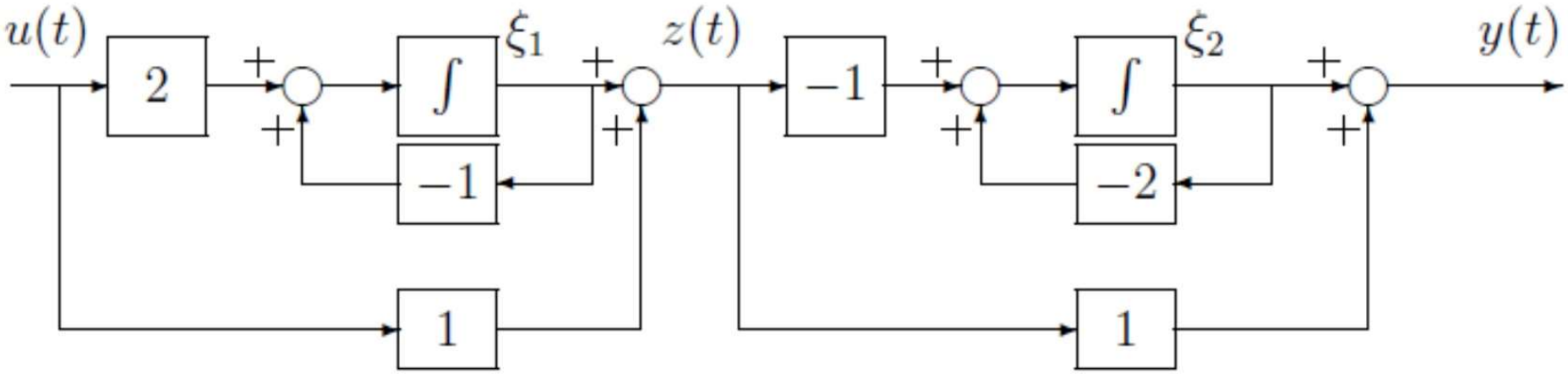
CW39

MAS508-G
Lab 6 – State-space Design: Controllability and Observability



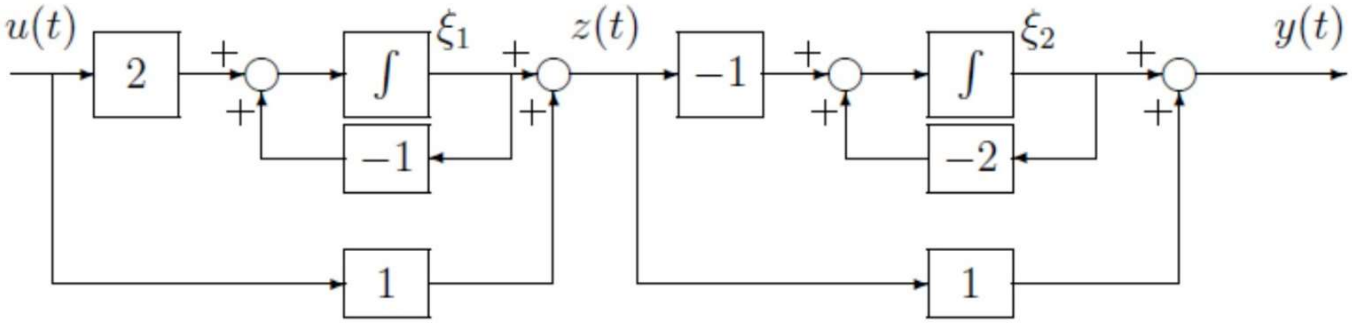
Example 1

- Consider the following system:



Task: System is controllable, but not fully observable. Prove it with controllability and observability tests, determining which eigenvalues cannot be observed.

Example 1– State-space model



$$\begin{cases} \dot{\xi}_1 = -\xi_1 + 2u \\ z = \xi_1 + u \\ \dot{\xi}_2 = -2\xi_2 - z \\ y = \xi_2 + z \end{cases} \Leftrightarrow \begin{cases} \dot{\xi}_1 = -\xi_1 + 2u \\ z = \xi_1 + u \\ \dot{\xi}_2 = -2\xi_2 - \xi_1 - u \\ y = \xi_2 + \xi_1 + u \end{cases}$$

$$\underset{\dot{x}}{\begin{bmatrix} \dot{\xi}_1 \\ \dot{\xi}_2 \end{bmatrix}} = \underset{A}{\begin{bmatrix} -1 & 0 \\ -1 & -2 \end{bmatrix}} \underset{x}{\begin{bmatrix} \xi_1 \\ \xi_2 \end{bmatrix}} + \underset{B}{\begin{bmatrix} 2 \\ -1 \end{bmatrix}} u$$

$$y = \underset{C}{\begin{bmatrix} 1 & 1 \end{bmatrix}} \underset{x}{\begin{bmatrix} \xi_1 \\ \xi_2 \end{bmatrix}} + \underset{D}{[1]} u$$

Example 1–Hautus criterion of controllability

Dynamic system $(\boldsymbol{A}, \boldsymbol{B})$ is exactly then **fully** controllable if, for all eigenvalues λ_i of the system matrix $\boldsymbol{A}$, it is valid:

$$\text{rank}[\lambda_i \boldsymbol{I} - \boldsymbol{A}, \boldsymbol{B}] = n$$

```
% State-space model, states x = [\xi_1 \xi_2]':
A = [-1 0; -1 -2];
B = [2; -1];
C = [1 1];
D = [];
S = ss(A,B,C,D);

% Controllability
n=size(A,2) %n=2
I=eye(2);
e=eig(A); %Eigenvalues of A: e=[-2; -1]
rank([e(1)*I-A,B]) %=2
rank([e(2)*I-A,B]) %=2
```

$\text{rank}[\lambda_1 \boldsymbol{I} - \boldsymbol{A}, \boldsymbol{B}] = n$ and $\text{rank}[\lambda_2 \boldsymbol{I} - \boldsymbol{A}, \boldsymbol{B}] = n \rightarrow$ system is fully controllable.

Example 1–Hautus criterion of observability

Dynamic system $(\boldsymbol{A}, \boldsymbol{C})$ is exactly then **fully** observable if, for all eigenvalues λ_i of the system matrix $\boldsymbol{A}$, it is valid:

$$\operatorname{rank}\begin{bmatrix}\lambda_i \boldsymbol{I} - \boldsymbol{A} \\ \boldsymbol{C}\end{bmatrix} = n$$

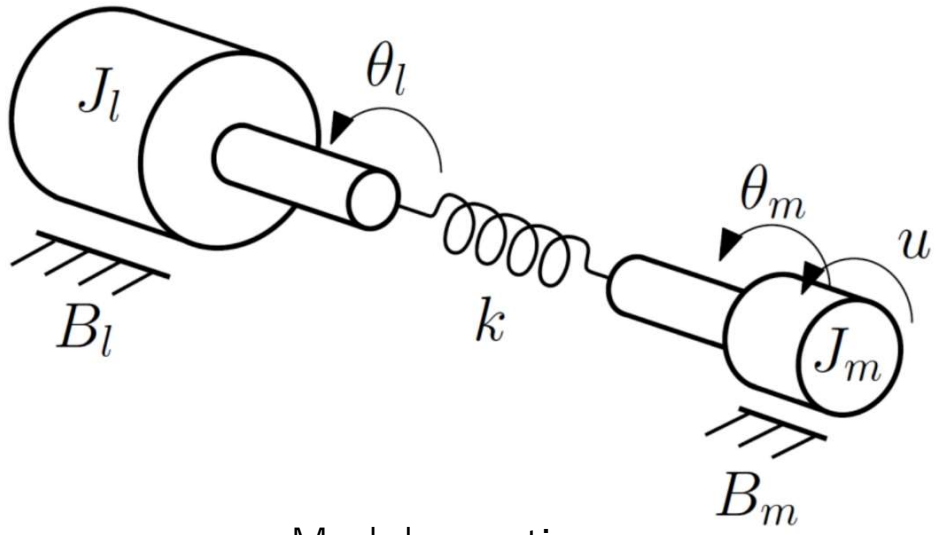
```
% Observability
rank([e(1)*I-A;C]) %=2
rank([e(2)*I-A;C]) %=1
```

$$\operatorname{rank}\begin{bmatrix}\lambda_1 \boldsymbol{I} - \boldsymbol{A} \\ \boldsymbol{C}\end{bmatrix} = n \text{ and } \operatorname{rank}\begin{bmatrix}\lambda_2 \boldsymbol{I} - \boldsymbol{A} \\ \boldsymbol{C}\end{bmatrix} \neq n$$

$\rightarrow$ Eigenvalue $\lambda_2 = -1$ cannot be observed

Example 2

- Consider the robot joint with elasticity 4th-order system




```
% Model parameters
Jm = 1; %Motor inertia
Jl=0.5; %Load inertia
Bm=10; %Motor damping
Bl=1; %Load damping
k=1000; %Stiffness
```

Model equations:

$$J_l\ddot{\theta}_l + B_l\dot{\theta}_l + k(\theta_l - \theta_m) = 0$$

$$J_m\ddot{\theta}_m + B_m\dot{\theta}_m - k(\theta_l - \theta_m) = u$$

Task: Derive state-space model and transform it into controllable canonical form by means of transformation matrix $\boldsymbol{T}$.

Example 2 – Derive state-space model

$$J_l\ddot{\theta}_l + B_l\dot{\theta}_l + k(\theta_l - \theta_m) = 0$$
$$J_m\ddot{\theta}_m + B_m\dot{\theta}_m - k(\theta_l - \theta_m) = u$$

reorganizing equations

$$\ddot{\theta}_l = -\frac{B_l}{J_l}\dot{\theta}_l - \frac{k}{J_l}\theta_l + \frac{k}{J_l}\theta_m$$
$$\ddot{\theta}_m = -\frac{B_m}{J_m}\dot{\theta}_m + \frac{k}{J_m}\theta_l - \frac{k}{J_m}\theta_m + \frac{1}{J_m}u$$

State-vector: $x = \begin{bmatrix}\theta_l & \dot{\theta}_l & \theta_m & \dot{\theta}_m\end{bmatrix}^T$
Output: $y = \theta_l$

Example 2 – Derive state-space model (cont.)

State-vector: $x = \begin{bmatrix} \theta_l & \dot{\theta}_l & \theta_m & \dot{\theta}_m \end{bmatrix}^T$

Output: $y = \theta_l$

$$\underbrace{\begin{bmatrix} \dot{\theta}_l \\ \ddot{\theta}_l \\ \dot{\theta}_m \\ \ddot{\theta}_m \end{bmatrix}}_{\dot{x}} = \underbrace{\begin{bmatrix} 0 & 1 & 0 & 0 \\ -\frac{k}{J_l} & -\frac{B_l}{J_l} & \frac{k}{J_l} & 0 \\ 0 & 0 & 0 & 1 \\ \frac{k}{J_m} & 0 & -\frac{k}{J_m} & -\frac{B_m}{J_m} \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} \theta_l \\ \dot{\theta}_l \\ \theta_m \\ \dot{\theta}_m \end{bmatrix}}_{x} + \underbrace{\begin{bmatrix} 0 \\ 0 \\ 0 \\ \frac{1}{J_m} \end{bmatrix}}_{B} u$$

$$y = \underbrace{\begin{bmatrix} 1 & 0 & 0 & 0 \end{bmatrix}}_{C} \underbrace{\begin{bmatrix} \theta_l \\ \dot{\theta}_l \\ \theta_m \\ \dot{\theta}_m \end{bmatrix}}_{x} + \underbrace{\begin{bmatrix} 0 \end{bmatrix}}_{D} u$$

Example 2 – Transformation matrix *T*

Compute transformation matrix T:

1) Controllability matrix: $Cr = [\boldsymbol{B}, \boldsymbol{AB}, ..., \boldsymbol{A}^{n-1}\boldsymbol{B}]$

```
Cr = ctrb(A,B);
```

2) Last row of Cr^{-1}: $c_1 = \begin{bmatrix} 0 & 0 & \dots & 1 \end{bmatrix} Cr^{-1}$

```
Cr_inv=inv(Cr);
c1=Cr_inv(end,:);
```

3) Collect elements into the inverse transformation matrix

$$T^{-1} = \begin{bmatrix} c_1 & c_1\boldsymbol{A} & c_1\boldsymbol{A}^2 & \dots & c_1\boldsymbol{A}^{n-1} \end{bmatrix}^T$$

```
T_inv = [c1; c1*A; c1*A^2; c1*A^3];
T=inv(T_inv);
```

Example 2 – Controllable canonical form

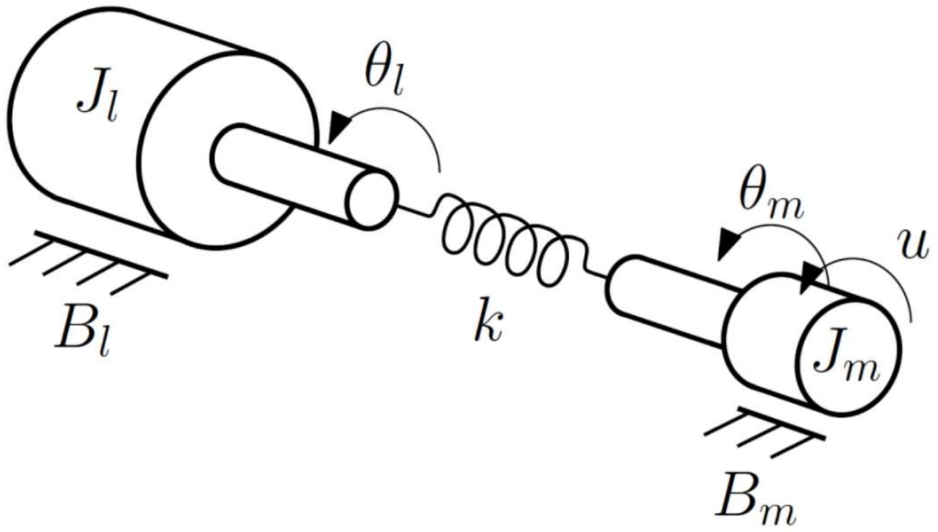
$$\boldsymbol{A}_C = \boldsymbol{T}^{-1}\boldsymbol{A}\boldsymbol{T}$$
$$\boldsymbol{B}_C = \boldsymbol{T}^{-1}\boldsymbol{B}$$
$$\boldsymbol{C}_C = \boldsymbol{C}\boldsymbol{T}$$
$$\boldsymbol{D}_C = \boldsymbol{D}$$

```
Ac = T^(-1)*A*T;
Bc = T^(-1)*B;
Cc = C*T;
Dc = D;
Scr = ss(Ac,Bc,Cc,Dc);
```

Note: original and transformed matrices describe one and the same system.

Example 3

- Consider the robot joint with elasticity (4th-order) system



Task: From the obtained controllable canonical form, derive the observable canonical form by using the duality principle.

Demonstrate that both canonical forms describe the same dynamic system (compare frequency responses (Bode diagram) and pole-zero map).

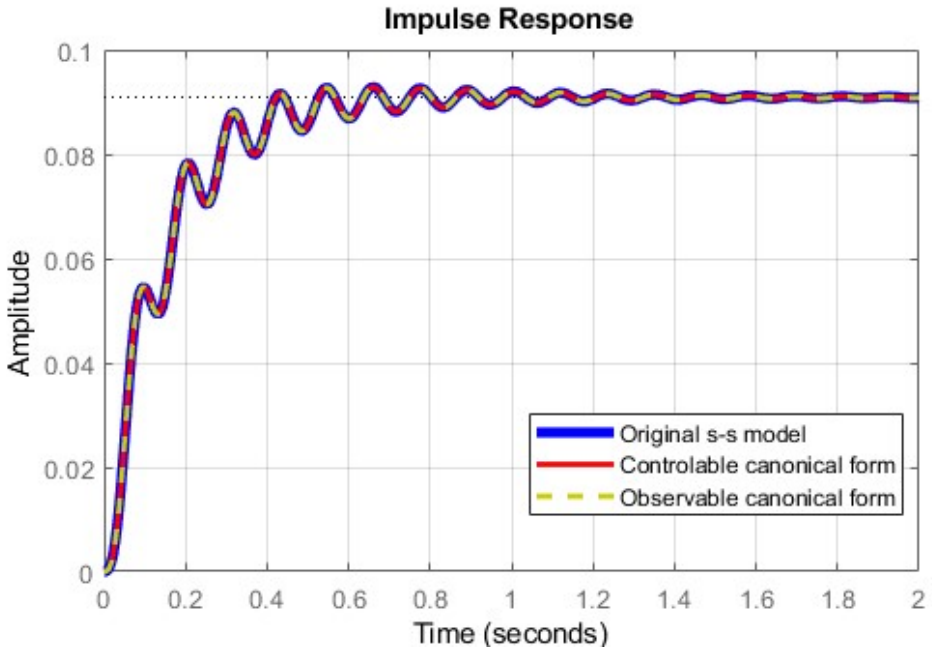
Example 3 – Observable canonical form

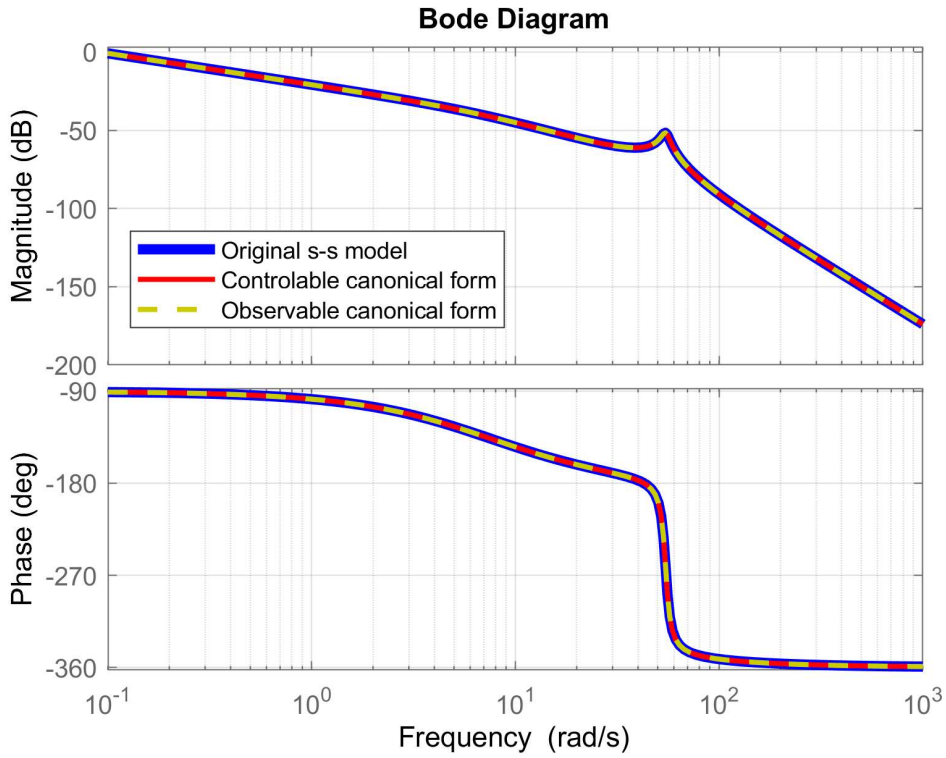
Controllable and observable canonical forms are dual to each other, i.e.

$$\boldsymbol{A}_O = \boldsymbol{A}_C^T$$
$$\boldsymbol{B}_O = \boldsymbol{C}_C^T$$
$$\boldsymbol{C}_O = \boldsymbol{B}_C^T$$
$$\boldsymbol{D}_O = \boldsymbol{D}$$

```
Ao = Ac';
Bo = Cc';
Co = Bc';
Do = Dc;
So = ss(Ao,Bo,Co,Do);
```

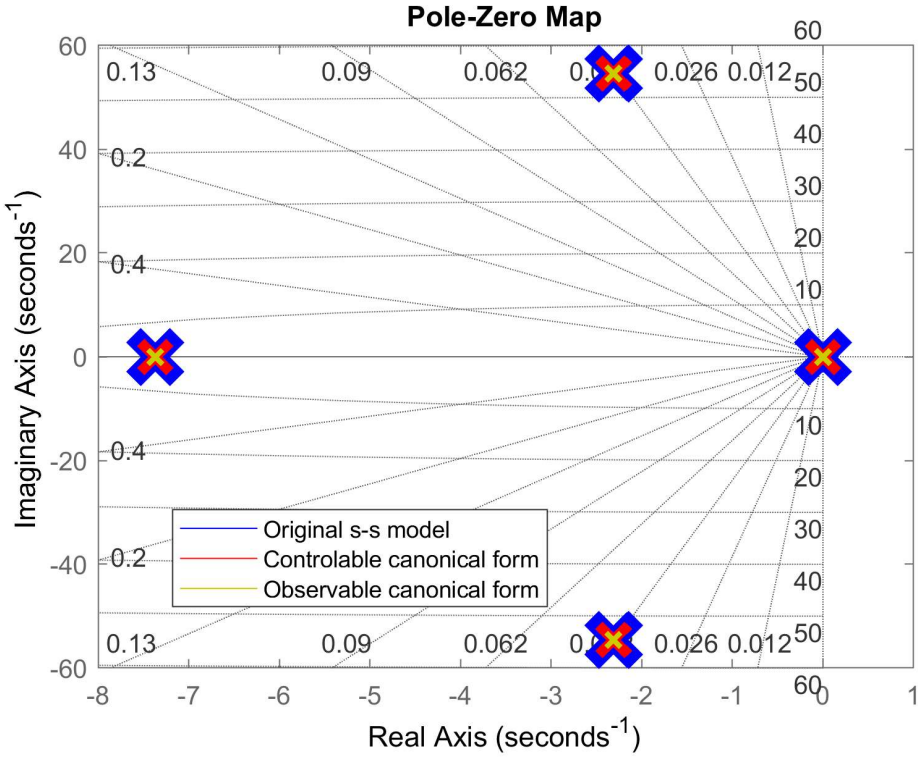
Example 3 – Compare both canonical forms





Both canonical forms describe one and the same system!

Example 3 – Compare both canonical forms (cont.)



pzmap() - creates a pole-zero plot of the dynamic system model sys
pzmap(sys)

```
impulse(S,'k',Sc,'g',So,'r',2); %Impulse response
bode(S,'k',Sc,'g',So,'r'); %Bode response
pzmap(S,'k',Sc,'g',So,'r'); %Pole-zero map
e_A=eig(A); e_Ac=eig(Ac); e_Ao=eig(Ao); %Compare eigenvalues
```

Extra Problems

Problem 1

Task: Show that the following system is controllable but not completely observable. Indicate which Eigenvalues do not meet the observability criteria.

$$\begin{aligned} \dot{x} &= Ax + Bu \\ y &= Cx \end{aligned} \tag{1}$$

where

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -6 & -11 & -6 \end{bmatrix}, B = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, C = \begin{bmatrix} 4 & 5 & 1 \end{bmatrix} \tag{2}$$

Problem 2

Task: Consider the transfer-function system

$$\frac{Y(s)}{U(s)} = \frac{25.04s + 5.008}{s^3 + 5.03247s^2 + 25.1026s + 5.008} \tag{3}$$

Using MATLAB, obtain a state-space representation of this system (hint: check tf2ss() function). Evaluate the controllability and observability of the system.